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Molecular Weight Determination of Block Copolymers by Pulsed Gradient Spin Echo NMR

Caroline Barrère,[†] Michaël Mazarin,[†] Rémi Giordanengo,[†] Trang N. T. Phan,[‡] André Thévand,[†] Stéphane Viel,[*,§] and Laurence Charles[†]

Laboratoire Chimie Provence, Spectrométries Appliquées à la Chimie Structurale, and Laboratoire Chimie Provence, Chimie Radicalaire, Organique et Polymères de Spécialité, Universités Aix-Marseille I, II & III - CNRS, UMR 6264, F-13397 Marseille Cedex 20, France, and Institut des Sciences Moléculaires de Marseille, Chimiométrie et Spectrométries, Aix-Marseille Université - CNRS, UMR 6263, F-13397 Marseille, France

Matrix-assisted laser desorption/ionization (MALDI) time-of-flight (TOF) mass spectrometry (MS) is the technique of choice to achieve molecular weight data for synthetic polymers. Because the success of a MALDI-MS analysis critically depends on a proper matrix and cation selection, which in turn relates closely to the polymer chemical nature and size, prior estimation of the polymer size range strongly helps in rationalizing MALDI sample preparation. We recently showed how pulsed gradient spin echo (PGSE) nuclear magnetic resonance could be used as an advantageous alternative to size exclusion chromatography, to rationalize MALDI sample preparation and confidently interpret MALDI mass spectra for homopolymers. Our aim here is to extend this methodology to the demanding case of amphiphilic block copolymers, for which obtaining prior estimates on the M_w values appears as an even more stringent prerequisite. Specifically, by studying poly(ethylene oxide) polystyrene block copolymers of distinct molecular weights and relative block weight fractions, we show how PGSE data can be used to derive the block M_w values. In contrast to homopolymers, such determination requires not only properly recorded calibration curves for each of the polymers constituting the block copolymers but also an appropriate hydrodynamic model to correctly interpret the diffusion data.

The ability of block copolymers to generate highly ordered nanostructures has been advantageously used in a wide range of applications, from advanced nanomaterials to biocompatible drug delivery systems.[1,2] In particular, amphiphilic copolymers have demonstrated numerous biological and chemical properties. The performance of such materials highly depends on structurally related parameters and requires both controlled polymerization techniques and powerful analytical methods to characterize key parameters such as molecular weight distribution and balance between hydrophobic and hydrophilic blocks.

Liquid state nuclear magnetic resonance (NMR) experiments are usually performed to determine the number average molecular weight (M_n) of each block. More precisely, the integrals of a properly selected set of ^{1}H resonances are compared and used to infer the M_n value of one of the block from the (typically known) M_n value of the other. Provided that the ^{1}H spectrum has been recorded using quantitative experimental parameters (i.e., enough time has been allowed for the resonances to relax completely) and no spectral overlapping is present, the estimation is rather accurate. Size exclusion chromatography (SEC) is also well-known for the common determination of polymer molar mass distribution.[3,4] Since SEC separates molecules based upon their hydrodynamic volume and not molecular weight, a conventional calibration using narrow standards of the same chemical nature and structure as the analyte is necessary. In case this requirement cannot be fulfilled, standards of a different nature from that of the polymer sample can, to some extent, be employed,[5] or one can use universal calibration,[6] which relies on Mark–Houwink–Kuhn–Sakurada (MHKS) parameters being known for the standards and the analyte. However, weak adsorption phenomena could affect polymer mass values as determined by SEC. Furthermore, in the case of block copolymers, the use of SEC to determine accurate molecular weight distribution faces two main limitations. On one hand, MHKS parameters depend on both the nature of comonomers and their relative proportion within the copolymer.[3] From the other hand, a multiple detection configuration is often required to reach reliable molecular weight data, since the intensity of the response obtained in each detection mode does not only depend on molecular weight but also on parameters which are a function of the macromolecule composition.[3,7] Mass spectrometry (MS) has become an increasingly important tool for polymer analysis since the development of soft ionization methods, such

* To whom correspondence should be addressed. Phone: (33) 491 288 900. Fax: (33) 491 282 897. E-mail: s.viel@univ-cezanne.fr.

† Laboratoire Chimie Provence, Spectrométries Appliquées à la Chimie Structurale.

‡ Laboratoire Chimie Provence, Chimie Radicalaire, Organique et Polymères de Spécialité.

§ Institut des Sciences Moléculaires de Marseille, Chimiométrie et Spectrométries.

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as matrix-assisted laser desorption/ionization mass spectrometry (MALDI). As oligomers can be detected as intact molecular adducts, MALDI MS spectra can be used to determine the number average (M_n) and weight average (M_w) molecular weights, as long as polymer polydispersity is not too high.[8−10] However, although successful MALDI analysis has been reported for various copolymers,[10−20] the case of amphiphilic block copolymers remains a challenge. Indeed, the choice of experimental conditions for sample preparation should be based on both the chemical nature of the comonomers and the size of each segment. To circumvent this issue, we recently developed a method consisting of hydrolyzing a targeted function in the junction moiety between the two blocks to produce two homopolymers which could then be independently mass-characterized using conventional MALDI protocols.[21] However, the scission of block copolymers into their constitutive segments can only be envisaged for those copolymers which contain a cleavable junction moiety and requires macromolecules to be chemically treated prior to mass analysis.

Alternatively, pulsed gradient spin echo (PGSE)[22−24] nuclear magnetic resonance (NMR) can be used to determine the M_w value of a synthetic polymer from the measurement of its molecular self-diffusion coefficient. Self-diffusion is defined in the liquid state as the random translational motion of molecules due to Brownian movement and is described by the self-diffusion coefficient D which depends on both the solvent viscosity and the molecular size.[25] In fact, similarly to SEC, the determination of M_w by PGSE is based on the hydrodynamic volume of the molecule and thus requires a calibration. However, the absence of a stationary phase is a clear advantage of PGSE over SEC as no adverse adsorption effect would be deplored. Moreover, as long as the contribution of the end groups to the hydrodynamic volume of a polymer chain is low, the use of standards having the same end groups as the analyte is not a requirement. Unlike MALDI-MS, PGSE can be applied to study samples spanning a wide range of molecular weights. Finally, PGSE does not require any specific sample preparation and is a rather fast technique that can easily be performed on generally available NMR instrumentation. PGSE has been applied to investigate fundamental aspects of polymer diffusion,[26,27] as well as to study polymer mixtures[28] and molecular weight distribution.[29−32] In particular, our group used M_w data as determined by PGSE to rationalize MALDI sample preparation and to confidently interpret the so-obtained mass spectra for poly(ethylene oxide) and poly(methyl methacrylate) homopolymers.[32] In this study, we propose to extend the use of PGSE to determine M_w parameters of diblock copolymers of the $(A)_n$-$(B)_m$ type, by focusing on a series of PEO-*b*-PS copolymers because of their numerous industrial applications.[33−42]

EXPERIMENTAL SECTION

Samples and Reagents. Poly(ethylene oxide) (PEO) standards (M_w = 320, 596, 1047, 4270, 5080, 25639, and 42700 g mol^{-1}) were supplied by Sigma Aldrich (St. Louis, MO) and polystyrene (PS) standards (M_w = 309, 972, 3460, 9830, 23800, and 74500 g mol^{-1}) were from Polymer Standards Service (Mainz, Germany). The deuterated solvent used in NMR experiments ($CDCl_3$, 99.9%) was from Euriso-Top (Saint-Aubin, France).

Copolymer Synthesis. Two series of SG1-capped PEO-*b*-PS block copolymers (Scheme 1) were synthesized from poly(ethylene oxide) methyl ether (PEO-Me) with $M_n \approx 2000$ and 5000 g mol^{-1} (Table 1). The synthetic procedure, already reported in a previous study[21] and briefly described here, involves three steps: the preparation of PEO-acrylate (**1**), the intermolecular radical addition of an alkoxyamine (hereafter referred to as MAMA-SG1) onto **1**, and the polymerization of styrene using the so-obtained

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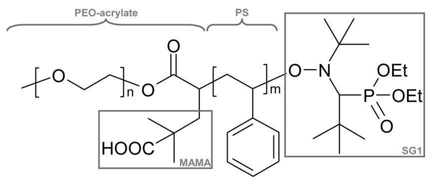
Scheme 1. Molecular Structure of the PEO-*b*-PS Block Copolymers



PEO-acrylate-MAMA-SG1 macroinitiator (**2**). The PEO-acrylate **1** was obtained by esterification of PEO-Me with an acryloil chloride. Addition of the MAMA-SG1 alkoxyamine to **1** occurs via a thermal homolysis of the C−ON bond, and the presence of activated olefin **1** leads to an intermolecular radical 1,2-addition. The so-formed PEO-acrylate-MAMA-SG1 **2** was then used to initiate the bulk polymerization of styrene at 90 °C during two days under inert atmosphere. The polymerization was stopped by quenching the reaction in an ice bath. The SG1-capped PEO-*b*-PS diblock copolymer was precipitated in cold *n*-hexane, rinsed with diethyl ether, and dried under vacuum at room temperature. The crude product was dissolved in dichloromethane and then poured into a large excess of diethyl ether or a mixture of diethyl ether with hexane or ethanol. The choice of nonsolvent for precipitation of the copolymer was made according to the ratio of PS block compared with PEO block. The copolymer was isolated by filtration and then dried under vacuum at room temperature to a constant weight.

PGSE Experiments. PGSE experiments were performed on a Bruker AVANCE spectrometer operating at 500 MHz for the ^{1}H Larmor frequency and using a 5-mm triple resonance inverse cryoprobe optimized for ^{1}H detection and equipped with an actively shielded *z*-gradient coil. The gradient coil was calibrated by measuring the diffusion coefficient of the residual proton in D_2O[24] and was found to be 55 G cm^{-1}. The temperature was set to 300 K and controlled with an air flow of about 545 L h^{-1} to avoid temperature fluctuations due to sample heating during the gradient pulses. $CDCl_3$ solutions were prepared by weighting an amount of sample directly into the NMR sample tube and adding 0.6 mL of deuterated solvent. For each sample, PGSE measurements were performed at different decreasing concentrations until reaching a constant D value within our ±3% experimental precision, indicating the dilution level was in the validity range of Flory's law (i.e., absence of obstruction or concentration effects on diffusion measurements). These repeated experiments increase the whole analysis time but are required for D measurement, hence M_w value, validation. Optimal concentration level was found to be 3.0 mg mL^{-1} for all samples. In these conditions, no significant difference in solution viscosity was noted.

The pulse sequences used for the measurements in $CDCl_3$ were based on a double-stimulated echo with bipolar gradients and a longitudinal eddy current delay (LED) used for minimizing spectral artifacts resulting from eddy currents.[29] Note that the double-stimulated echo sequence was used to avoid artifacts caused by thermal convection.[28] In these pulse sequences, the amplitude of a NMR resonance observed at the echo is given by[22,23]

$$I = I_0 \exp(-D(\gamma g\delta)^2(\Delta - \varepsilon(\delta)))$$

where I_0 is the resonance amplitude at zero gradient strength, γ is the magnetogyric ratio of the observed nucleus, g and δ are the strength and the duration of the gradient pulses, respectively, and Δ is the diffusion time, that is, the time during which the diffusion is monitored. $\varepsilon(\delta)$ is a correction factor that depends on both δ and the pulse sequence. Usually, only the gradient strength is varied, and all delays are kept constant to avoid any complication arising from magnetic relaxation. Note that, in large polymers, spin relaxation rates are typically determined by segmental motion (local) rather than overall tumbling rates, and hence relaxation times are usually assumed to be independent of molecular weight.[29] Specifically, the gradient strength was quadratically incremented in 16 steps from 6 to 95% of its maximum value. The diffusion time was set to 500 ms, and gradient pulse durations were optimized for each experiment to achieve at the largest gradient amplitude a decrease in the resonance intensity larger than 97%; typically, bipolar sine gradient pulses between 1.0 and 2.3 ms were employed. The gradient pulse recovery time and the longitudinal eddy current delay were set to 0.15 and 25 ms, respectively. On average a PGSE experiment lasted about 47 min (32 scans for each gradient value), but recent development of this technique allows further time saving to be envisioned in the future.[43] After Fourier transformation and phase correction, the baseline of the spectra was carefully adjusted. The data were

Table 1. Copolymer Distribution Parameters Determined by MALDI MS (after Copolymer Hydrolysis)[21] and Quantitative ^{1}H NMR

	MALDI MS						^{1}H NMR		
sample no.	M_n^{PEO} (g mol^{-1})	M_n^{PS} (g mol^{-1})	M_n^{T} (g mol^{-1})^a	M_w^{PEO} (g mol^{-1})	M_w^{PS} (g mol^{-1})	M_w^{T} (g mol^{-1})^a	M_n^{PEO} (g mol^{-1})^b	M_n^{PS} (g mol^{-1})^c	M_n^{T} (g mol^{-1})^a
A1	1835	4815	7117	1873	5385	7725	1910	4960	7337
A2	1859	14563	16889	1909	15360	17736	1950	13539	15956
A3	1829	16062	18358	1872	16921	19260	1980	17594	20041
A4	1826	20571	22864	1868	21647	23982	1910	21375	23752
B1	4967	3699	9133	4994	3983	9444	4770	3700	8937
B2	4981	4250	9698	5104	4332	9903	4660	4960	10087
B3	4959	9016	14442	5138	9080	14685	4540	9200	14207
B4	4962	13243	18672	5022	14694	20183	4620	13350	18437
B5	4969	21697	27133	5017	22715	28199	4610	21300	26377

^{a} Calculated by adding the mass of the junction moiety and of the end groups (467 g mol^{-1}) to the mass of the PEO and PS blocks. ^{b} Calculated considering the integral of the terminal OCH_3 group as a reference. ^{c} Calculated considering the M_n^{PEO} value as a reference.

Table 2. Total Weight Average Molecular Weight (M_w^T) Values for the Copolymers under Study As Obtained from PGSE Self-Diffusion Coefficient Measurements and Using the Homopolymer Calibration Curves[a]

	PGSE	PEO calibration curve		PS calibration curve	
no.	D ($\times 10^{-10}$ m^2 s^{-1})	M_w^T (g mol^{-1})	dev. (%)[b]	M_w^T (g mol^{-1})	dev. (%)[b]
A1	1.474	7874	+2	13581	+76
A2	1.135	12276	−31	21770	+23
A3	1.077	13421	−30	23932	+24
A4	0.976	15865	−34	28588	+19
B1	1.384	8764	−7	15217	+61
B2	1.256	10335	+4	18131	+83
B3	1.102	12907	−12	22961	+56
B4	0.948	16669	−17	30131	+49
B5	0.875	19101	−32	34821	+23

[a] Deviations are calculated with respect to data derived from MALDI analysis. [b] Taking MALDI-MS values as a reference, see Table 1.

analyzed by plotting the signal intensities (areas) as a function of the gradient strength and fitting the resulting decays with a nonlinear least-squares fit. This simple data analysis scheme was motivated by the relatively narrow molecular weight distribution of the samples obtained by MALDI (see Table 1). However, inconsistent results may be obtained in the case of significant polymer polydispersity, as emphasized in refs 29 and 30.

RESULTS

Initially, the size of each block for all the samples of both PEO-*b*-PS block copolymer series, annotated A_i ($M_n^{PEO} \approx 2000$ g mol^{-1}) and B_i ($M_n^{PEO} \approx 5000$ g mol^{-1}), respectively, was determined by ^{1}H NMR analysis and by MALDI MS after hydrolysis of the studied copolymers.[21] Adding the mass of both the junction moiety and the end-groups to these values (i.e., 467 g mol^{-1}), one could obtain distribution parameters of the whole macromolecules, M_n^T and M_w^T, as reported in Table 1. The MALDI data used as a reference indicate that the analyzed copolymers are characterized by a relatively narrow molecular weight distribution (polydispersity index < 1.1). This situation is rather typical for samples obtained by NMP, and it helps simplifying the PGSE data analysis by minimizing the artifacts arising from sample polydispersity.[29,30]

PGSE experiments were then performed to measure the self-diffusion coefficient of each PEO-*b*-PS copolymer (Table 2). The diffusion of polymers is well understood and has been extensively studied.[44] The M_w parameter of a homopolymer can be related to its self-diffusion coefficient D, as measured in very dilute solutions, according to Flory's law[44]

$$D = KM_w^{-\alpha} \quad (1)$$

where K and α are scaling parameters that depend on the polymer structure as well as experimental conditions (solvent viscosity and temperature). Typically, these scaling parameters are obtained by analyzing a series of monodisperse polymer standards under a given set of experimental conditions. In a first approximation, two calibration curves were established by analyzing a set of PEO and PS homopolymer standards, respectively, to express copolymer M_w^T values in PEO or PS equivalents. By fitting the data to the Flory's law, calibration curves were found to be such as $D = 2.890 \times 10^{-8} M_w^{-0.5885}$ for PEO and $D = 2.869 \times 10^{-8} M_w^{-0.5539}$ for PS (Figure 1). From this figure, it could already be anticipated that, depending on the considered calibration curve, different M_w^T values would be obtained for each copolymer (Table 2). Using the PS calibration curve, calculated deviations are much greater than the generally accepted error of 10% in PGSE.[32] However, it should be noted that, for macromolecules with PS and PEO blocks of similar size, typically in samples A_1, B_1, and B_2, copolymer M_w^T values as expressed in PEO equivalent are consistent with the expected data. These results suggest that, as long as the PS block size is not too large as compared to the PEO segment, a PEO-*b*-PS copolymer would behave like a PEO homopolymer in terms of diffusion. Nevertheless, apart from these three particular cases, the hydrodynamic model used for homopolymer diffusion clearly does not apply for block copolymers.

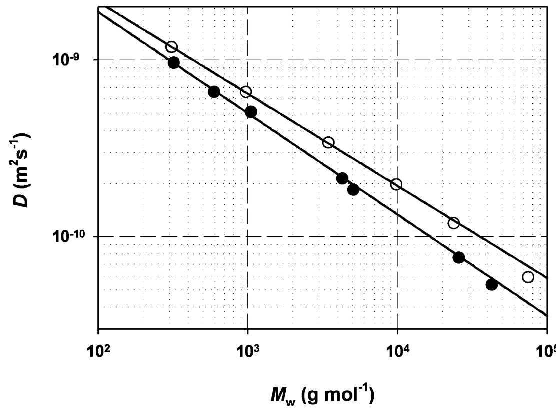


Figure 1. Double-logarithmic plot showing the evolution of the self-diffusion coefficient D as a function of the weight average molecular weight M_w for a series of seven PEO standards (black circles) and a series of six PS standards (white circles). The D values were measured by PGSE experiments recorded at 300 K in dilute $CDCl_3$ solutions. In both cases the straight lines are best fit to eq 1 and allow the scaling parameters for PEO and PS to be determined.

Many authors have investigated the conformation of block copolymers in dilute solution, using mainly dynamic light scattering (DLS) and small angle neutron scattering (SANS) experiments. These investigations have shown that the conformation of a diblock copolymer depends in a complicated way on the nature of the blocks and the solvent. Overall, two main cases can be envisaged depending on the interactions between the two blocks: a conformation in which the two segments behave independently of each other (hereafter referred to as segregated conformation) and a conformation involving multiple contacts between the two blocks.[45] For instance, Prud'homme and Bywater studied poly(styrene-*b*-isoprene) diblock copolymers by DLS and showed that the two types of chain segments must be separated in solution, especially in poor solvent for the block.[46] In contrast, Ionescu et al. studied a similar system by SANS but found no evidence for block segregation in a good solvent for the two

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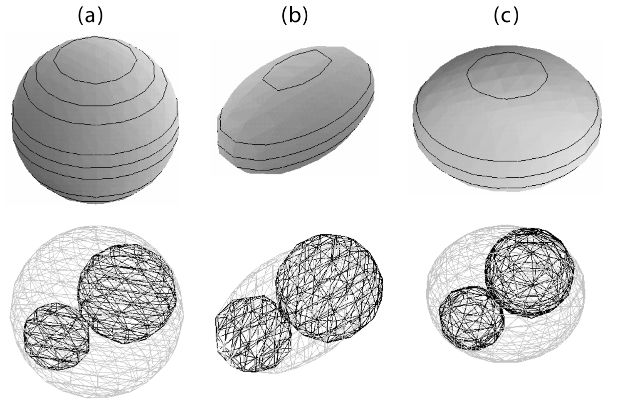

Figure 2. Illustration of the three hydrodynamic models considered in this study. In all cases a segregated conformation was assumed for the copolymers. In other words, the copolymer blocks are supposed to behave in solution independently from one another. Each block is thus considered as a random coil with a spherical conformation. In this case, the resulting copolymer shape can be modeled as (a) a sphere or as an ellipsoid of revolution being (b) prolate [cigar-like] or (c) oblate [disk-like]. The three cases are depicted here by using copolymer A4 as an example, for which the PS block is much larger than the PEO block.

blocks,[47,48] whereas Han and Mozer arrived at the opposite conclusion while using a similar technique on the same system.[49] More specifically, in the case of amphiphilic copolymers, Tanaka et al. investigated poly(styrene-*b*-methyl methacrylate) copolymers in 2-butanone by DLS and showed that their results were not consistent with a segregated conformation.[50] On the contrary, by studying a poly(styrene-*b*-2-vinylpyridine) block copolymer with a composition of 50:50 in dilute solution by SANS measurements, Matsushita et al. more recently showed that the conformation of one block chain within a diblock copolymer was not affected by the presence of the other block, regardless of the solvent power, as long as the molecular weight was not too high (M_n < 84 000 g mol^{-1}).[51] In view of the strong amphiphilic character of the PEO-*b*-PS copolymers studied here, a similar segregated conformation with no contact between the two blocks was assumed, considering the entire copolymer macromolecular chain as a single random coil.

The simplest hydrodynamic pattern to account for this conformation is the spherical model, where the copolymer hydrodynamic radius, r_H^{copo}, is the sum of the two block hydrodynamic radii, r_H^{PEO} and r_H^{PS} (Figure 2a). Knowing each block weight average molecular weight, we can express their radius using Flory's law (eq 1) and the Stokes−Einstein equation for a spherical particle of colloidal dimension in a fluid continuum, such as

$$D = \frac{kT}{6\pi\eta r_H} \quad (2)$$

where k is Boltzmann's constant, T the absolute temperature, and η the fluid viscosity. Combining these two equations, hydrodynamic radii of each block and of the copolymer macromolecule in the spherical model are given by

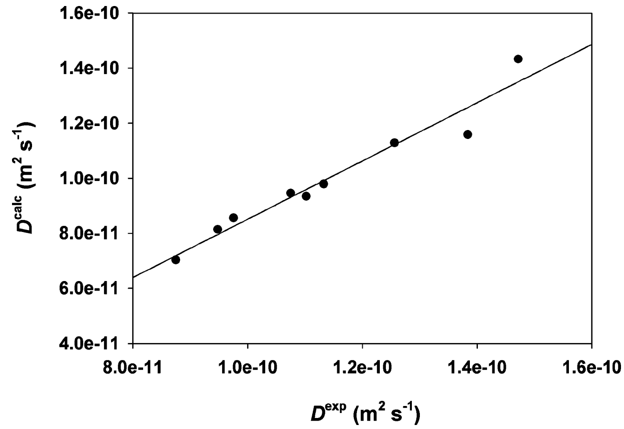

Figure 3. Comparison between copolymer self-diffusion coefficients measured experimentally (D^{exp}) and copolymer self-diffusion coefficients calculated (D^{calc}) by assuming a segregated conformation and a spherical hydrodynamic model (see text). A linear regression to the data gave $D^{calc} = 0.8808D^{exp}$ ($R^2 = 0.9226$), indicating that the D^{calc} values are underestimated.

$$r_H^{PEO} = \frac{kT}{6\pi\eta K^{PEO}(M_w^{PEO})^{-\alpha_{PEO}}} \quad \text{and} \quad r_H^{PS} = \frac{kT}{6\pi\eta K^{PS}(M_w^{PS})^{-\alpha_{PS}}} \quad (3)$$

$$r_H^{copo} = r_H^{PEO} + r_H^{PS} = \frac{kT}{6\pi\eta K^{PEO}(M_w^{PEO})^{-\alpha_{PEO}}} + \frac{kT}{6\pi\eta K^{PS}(M_w^{PS})^{-\alpha_{PS}}} \quad (4)$$

According to eq 2, the copolymer diffusion coefficient, D^{copo}, can be written as

$$D^{copo} = \frac{K^{PEO}(M_w^{PEO})^{-\alpha_{PEO}}K^{PS}(M_w^{PS})^{-\alpha_{PS}}}{K^{PEO}(M_w^{PEO})^{-\alpha_{PEO}} + K^{PS}(M_w^{PS})^{-\alpha_{PS}}} \quad (5)$$

To calculate D^{copo} from eq 5, M_w^{PEO} and M_w^{PS} values measured for each block by MALDI MS (Table 1) were used, whereas the K and α scaling parameters for PS and PEO were obtained from the respective homopolymer calibration curves shown in Figure 1. For each copolymer sample, the so-calculated D^{copo} value was subsequently compared to the experimental self-diffusion coefficient, and a correlation coefficient of 0.88 was obtained (Figure 3). This indicated that the calculated self-diffusion coefficients were underestimated, suggesting that the spherical hydrodynamic model was inadequate. In particular, this model does not take into account the shape of the diffusing molecules. The previously used eq 2 is actually a simplified law which applies in the particular case of spherical particles of colloidal dimension in a fluid continuum. The general Stokes−Einstein diffusion law is

$$D = \frac{kT}{c(r_{solvent}, r_H)f_s\pi\eta r_H} \quad (6)$$

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where $c(r_{solvent}, r_H)$ is a coefficient that depends on the ratio between the hydrodynamic radius of the solvent, $r_{solvent}$, and that of the particle, r_H, whereas f_s is a correction factor that takes into account the shape of the molecule. When the radius of the molecule is at least five times larger than that of the solvent (which applies in our case where PEO-*b*-PS copolymers diffuse in $CDCl_3$), the $c(r_{solvent}, r_H)$ coefficient equals 6.[52] The shape of the molecule was studied with respect to the theoretical models developed by Perrin[53] for the diffusion of prolate and oblate ellipsoidal molecules, as illustrated in Figure 2b,c, respectively. When considering ellipsoids of revolution, it is convenient to define two semiaxes a and b, which are the revolution and equatorial semiaxis, respectively, as well as the ρ ratio defined as $\rho = b/a$. Accordingly, for a prolate ellipsoidal molecule (Figure 2b), a is equal to the sum $r_H^{PEO} + r_H^{PS}$, whereas b is the hydrodynamic radius of the biggest block, $r_H^{biggest}$. In contrast, for an oblate ellipsoidal molecule (Figure 2c), a is equal to $r_H^{biggest}$ whereas b is the sum of the block hydrodynamic radii. Perrin found that the self-diffusion coefficient, D, of an ellipsoidal molecule differs from the self-diffusion coefficient, D_0, of a spherical molecule having the same volume by a coefficient, hereafter referred to as $f_s(\rho)$, calculated for the prolate and oblate models according to

$$\frac{1}{f_s^{prolate}(\rho)} = \frac{D^{prolate}}{D_0^{prolate}} = \frac{\rho^{2/3}}{\sqrt{1-\rho^2}} \ln\left(\frac{1+\sqrt{1-\rho^2}}{\rho}\right) \tag{7a}$$

$$\frac{1}{f_s^{oblate}(\rho)} = \frac{D^{oblate}}{D_0^{oblate}} = \frac{\rho^{2/3}}{\sqrt{\rho^2-1}} \arctan(\sqrt{\rho^2-1}) \tag{7b}$$

Combining Flory's law for PEO and PS with the Stokes–Einstein equation (eq 6) and using the denomination "biggest" and "smallest" in subscript to refer to the biggest and the smallest blocks, respectively, eqs 8a and 8b were obtained for the prolate and oblate ellipsoids, respectively:

$$\rho^{prolate} = \frac{b}{a} = \frac{r_H^{biggest}}{r_H^{PEO} + r_H^{PS}} = \frac{K^{smallest}(X^{smallest})^{-\alpha_{smallest}}(M_w^T)^{-\alpha_{smallest}}}{K^{smallest}(X^{smallest})^{-\alpha_{smallest}}(M_w^T)^{-\alpha_{smallest}} + K^{biggest}(X^{biggest})^{-\alpha_{biggest}}(M_w^T)^{-\alpha_{biggest}}} \tag{8a}$$

$$\rho^{oblate} = \frac{b}{a} = \frac{r_H^{PEO} + r_H^{PS}}{r_H^{biggest}} = 1 + \frac{K^{biggest}(X^{biggest})^{-\alpha_{biggest}}}{K^{smallest}(X^{smallest})^{-\alpha_{smallest}}}(M_w^T)^{-\alpha_{biggest}+\alpha_{smallest}} \tag{8b}$$

where $X^{biggest}$ and $X^{smallest}$ are the weight fractions of the biggest and the smallest blocks, respectively.

Perrin's shape correction coefficients, $f_s^{prolate}(\rho)$ and $f_s^{oblate}(\rho)$, can be calculated for both models using eqs 7a and 8a and eqs 7b and 8b, respectively, to be further used to determine the respective $D^{prolate}$ and D^{oblate} values, provided that the corresponding D_0 values are known. The radius $r_H{}^0$ of the sphere with the same volume as the ellipsoid of revolution is expressed by

$$r_H^0 = a\rho^{2/3} \tag{9}$$

Therefore, using the Stokes–Einstein equation for a sphere and Flory's law, $r_H{}^0$ and D_0 can be calculated for the prolate and the oblate ellipsoids, respectively, such as

$$(r_H^0)^{prolate} = \frac{kT}{6\pi\eta}\left(\frac{1}{K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}}} + \frac{1}{K^{smallest}(X^{smallest}M_w^T)^{-\alpha_{smallest}}}\right)\rho^{2/3} \tag{10a}$$

$$(r_H^0)^{oblate} = \frac{kT}{6\pi\eta K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}}}\rho^{2/3} \tag{10b}$$

$$D_0^{prolate} = \frac{K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}}K^{smallest}(X^{smallest}M_w^T)^{-\alpha_{smallest}}}{K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}} + K^{smallest}(X^{smallest}M_w^T)^{-\alpha_{smallest}}}\rho^{-2/3} \tag{11a}$$

$$D_0^{oblate} = K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}}\rho^{-2/3} \tag{11b}$$

Finally, combining eqs 7a, 8a, and 11a, a theoretical expression can be obtained for the copolymer self-diffusion coefficient according to each model:

$$D^{prolate} = \frac{1}{\sqrt{1-\rho^2}}\ln\left(\frac{1+\sqrt{1-\rho^2}}{\rho}\right) \times \frac{K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}}K^{smallest}(X^{smallest}M_w^T)^{-\alpha_{smallest}}}{K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}} + K^{smallest}(X^{smallest}M_w^T)^{-\alpha_{smallest}}} \tag{12a}$$

$$D^{oblate} = \frac{1}{\sqrt{\rho^2-1}}\arctan(\sqrt{\rho^2-1})K^{biggest}(X^{biggest}M_w^T)^{-\alpha_{biggest}} \tag{12b}$$

The weight fraction of the biggest and the smallest blocks can be determined from M_n values obtained after signal integration in quantitative ^{1}H NMR spectra. However, since the $\alpha_{biggest}$ and $\alpha_{smallest}$ parameters used as exponents are noninteger values, eqs 12a and 12b cannot be solved exactly. Nevertheless, M_w^T can be reached by considering and drawing the following two functions:

$$f^{prolate}(M) = D^{prolate} - D_{copo}^{exp} \tag{13a}$$

$$f^{oblate}(M) = D^{oblate} - D_{copo}^{exp} \tag{13b}$$

which should intercept the x axis (i.e., $f(M) = 0$) at $M = M_w^T$. Such a graphical strategy allows the copolymer M_w^T value to be straightforwardly determined, without using expensive and

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Table 3. Total Weight Average Molecular Weight (M_w^T) Values for the Copolymers under Study Calculated Using the Copolymer Self-Diffusion Coefficients, As Measured by PGSE, and the Prolate and Oblate Ellipsoidal Hydrodynamic Models[a]

	prolate model		oblate model	
no.	M_w^T (g mol^{-1})	dev. (%)[b]	M_w^T (g mol^{-1})	dev. (%)[b]
A1	12658	+64	9041	+17
A2	24489	+38	15762	−11
A3	28261	+47	17780	−8
A4	35190	+47	21681	−10
B1	12159	+29	8481	−10
B2	14799	+49	10599	+7
B3	20327	+38	14860	+1
B4	28297	+40	19840	−2
B5	35880	+27	23910	−15

[a] Deviations are calculated with respect to data derived from MALDI analysis. [b] Taking MALDI-MS values as a reference, see Table 1.

complex calculation programs. The weight average molecular weights of the studied copolymers obtained using both ellipsoidal models are reported in Table 3, together with the deviation with respect to the MALDI MS data used as a reference. While the prolate model is obviously not adequate to describe the studied block copolymer conformation (relative deviations are above 27% for all samples), the oblate model gave rise to much more accurate M_w^T values, with errors below 10% in most cases, showing the validity of this hydrodynamic model for the studied PEO-*b*-PS copolymers.

Owing to the polymerization process used to synthesize the studied PEO-*b*-PS copolymers, starting from a well-characterized PEO homopolymer standard, the M_w parameter can be easily obtained for the PS segment once the copolymer M_w^T value has been graphically determined. In case the PEO block size is unknown, M_w parameters can still be obtained for each block using the following graphical procedure. Combining eq 8b with the expression of $M_w^T = M_w^{\text{smallest}} + M_w^{\text{biggest}} + 467$, a $g(M)$ function can be written as follows

$$g(M) = (A^{-1/\alpha_{\text{biggest}}})(M^{\alpha_{\text{smallest}}/\alpha_{\text{biggest}}}) + M + 467 - M_w^T \tag{14}$$

with $A = K^{\text{smallest}}(\rho - 1)/K^{\text{biggest}}$, which is equal to zero when $M = M_w^{\text{smallest}}$.

The so-obtained results for block weight average molecular weights are presented in Table 4. These data indicate that block M_w values, as determined from the oblate ellipsoidal model, compare pretty well with data obtained after MALDI MS of homopolymers produced from copolymer hydrolysis. For PEO, absolute errors are below 17% with an average value of about 8%. Larger errors were obtained for the PS block with an average value of about 12.5% and deviation up to 20−30% for two of the nine studied copolymer samples. It should be noted that the size of the errors, for both the whole macromolecule and the individual segments, could not be related to the relative size of the blocks within the copolymer.

Table 4. Copolymer Block Size (M_w^{PEO} and M_w^{PS}) Calculated Using the Copolymer Self-Diffusion Coefficients, As Measured by PGSE and the Oblate Ellipsoidal Hydrodynamic Model[a]

no.	M_w^{PEO} (g mol^{-1})	dev. (%)[b]	M_w^{PS} (g mol^{-1})	dev. (%)[b]
A1	2562	+37	6012	+12
A2	2011	+5	13284	−14
A3	1799	−4	15514	−8
A4	1782	−5	19432	−10
B1	4304	−14	3710	−7
B2	4873	−5	5259	+21
B3	4887	−5	9506	+5
B4	5106	+2	14267	−3
B5	4235	−16	19208	−15

[a] Deviations are calculated with respect to MALDI data. [b] Taking MALDI-MS values as a reference, see Table 1.

CONCLUSION

This study has shown that the conformation in solution of the PEO-*b*-PS block copolymers studied here is well described by a segregated conformation as long as a proper model is used to account for the hydrodynamic shape of the copolymers in solution. Among the three tested hydrodynamic models, the oblate ellipsoidal pattern was found to adequately fit the copolymer self-diffusion coefficient data, as experimentally measured by PGSE, and allowed their M_w values to be accurately determined. Specifically, a mathematical expression based on Flory's law and the Stokes−Einstein's equation was derived to give the copolymer self-diffusion coefficient. This expression involved a series of scaling parameters (as determined by PGSE for homopolymers of the same nature as the constitutive blocks), the weight fraction of each segment in the copolymer (as calculated from the number average molecular weights obtained for each block after signal integration in quantitative ^{1}H NMR experiments), and the M_w^T parameter of the macromolecule. Once copolymer self-diffusion coefficients were experimentally measured by PGSE, this mathematical expression could be graphically solved to reach the copolymer weight average molecular weight, within the generally accepted 10% relative error in most of the studied cases. While the methodology proposed here has mainly focused on diblock $(A)_n$-$(B)_m$ copolymers of relatively narrow molecular weight distribution, it could be of interest for studying other types of copolymers, and investigations along those lines are currently in progress.

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